## 18.1 Answers/Solutions

1. **Probability Rules and Axioms.** Make a list of all the probability rules and axioms we have learned. List them by name and using good probability notation.

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***Solution.***
You can find these on past sheets. Here's an example:
Sum Rule: The probability of the union of mutually exclusive events is the sum of their individual probabilities.

- $p(A \cup B) = p(A) + p(B)$, provided $p(A \cap B) = 0$.

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2. **Mutually exclusive vs independent.**

   a. What does it mean for events $A$ and $B$ to be **mutually exclusive**? Express this in words an in probility notation.
   b. What does it mean for events $A$ and $B$ to be **independent**? Express this in words an in probility notation.
   c. Is it possible for events $A$ and $B$ to be mutually exclusive and independent? Give an example where this happens or explain why it cannot happen.
   d. Give an example of a probability rule that only holds if events are mutually exclusive.
   e. Give an example of a probability rule that only holds if events are independent.

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***Solution.***

a. Events $A$ and $B$ are **mutually exclusive** if they cannot both happen at the same "time" (i.e., on the same repetition of the random process). $p(A \cap B) = 0$.

b. Events $A$ and $B$ are **independent** if knowing whether one of them ocurred does not change the probability of the other: $p(A \mid B) = p(A)$.

c. If $A$ and $B$ are mutually exclusive and independent, then $p(A) = p(A \mid B) = 0 = p(B \mid A) = p(B)$. But if $p(A) = p(B) = 0$, then the conditional probabilities are not even defined. So two events cannot be both independent and mutually exclusive.

d. The sum rule above requires mutually exclusive events. (There is a generalized version that holds for any events and has a subtracted term.)

e. If $A$ and $B$ are independent, then $p(A \,\text{and}\, B) = p(A) \cdot p(B)$.

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3. a. What is the difference between $p(A \mid B)$ and $p(B \mid A)$?
   b. Give an example where these are $p(A \mid B) = p(B \mid A)$.
   c. Give a different example where $p(A \mid B) \neq p(B \mid A)$.
   d. True or False: If $A$ and $B$ are independent, then $p(A \mid B) = p(B \mid A)$? (Either explain why this is always true or give an example where it is false.)

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***Solution.***

a. $p(A \mid B)$ asks what fraction of the time that $B$ occurs will $A$ also occur. $p(B \mid A)$ asks what fraction of the time that $A$ occurs will $B$ also occur.

b. Flip two coins. Let $A$ be the event that the first is heads. Let $B$ be the event that the second is heads. Then $p(A \mid B) = p(B \mid A) = \frac{1}{2}$.

c. Flip a coin and roll a die. Let $A$ be the event that the coin is heads. Let $B$ be the event that the die produces a 6. $p(A\ midB) = p(A) = \frac{1}{2}$. $p(B \mid A) = \frac{1}{6}$.

d. False. See the example in part c. Those events are independent.

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4. **Binomial distributions**

   a. For random variable to be a binomial random variable, what 5 things must be true?
   b. Give an example of a situation that is $\mathsf{Binom}(20, 0.25)$.

**Solution.**

a. These five things are listed on worksheet 15.
b. Roll a four-sided die 20 times and count the number of 3's that are rolled.

5. Suppose $X \sim \mathsf{Binom}(20, 0.25)$. (a) What is $p(X = 4)$? (b) What is $\mathrm{E}(X)$?

**Solution.**

a. $p(X = 4) = \binom{20}{4}(0.25)^4(0.75)^{16} = 0.1896855$
b. $\mathrm{E}(X) = 20 \cdot 0.25 = 5$?

6. Suppose at a certain college 60% of students are female, 10% of female students are from international students, and 12% of male students are international. What percentage of the international students are female?

   a. Do you expect the answer to be less than 12%, between 12% and 60%, or over 60%? Explain.
   b. Express the three numbers in this problem and the number asked for in the question using good probability notation.
   c. Answer the answer numerically.

**Solution.**
Let $F$ be the event that a student is female. Let $I$ be the event that a student is international. $0.6 = p(F)$. $0.10 = p(I \mid F)$. $0.12 = p(I \mid F^c)$. We are asked to compute $p(W \mid I)$.

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0.6 * 0.1 / (0.6 * 0.1 + 0.4 * 0.12)
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[1] 0.5555556
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7. Suppose that we flip a fair coin until either it comes up tails twice or we have flipped it six times.

   a. Is this a binomial situation? Explain.
   b. What is the expected number of times we flip the coin?

**Solution.**
This is not binomial. We don't know the number of trials in advance.

```
vals  2.00000 3.0000000 4.0000000 5.00000000 6.0000000
probs 0.09375 0.1054687 0.1054687 0.09887695 0.5964355
prods 0.18750 0.3164062 0.4218750 0.49438477 3.5786133
```

Expected value is 4.9987793.

8. Suppose that we flip a fair coin until it comes up tails twice, no matter how many tries that takes.

   a. Is this a binomial situation? Explain.
   b. What is the expected number of times we flip the coin?

**Solution.**
Still not binomial (we still don't know the number of trials in advance).
Part b is challenging. If $X$ is our random variable, we can split it up into two parts $X = X_1 + X_2$ where $X_1$ is the number of flips until we get the first tail and $X_2$ is the number of flips after the first tail until we get the second. It is pretty easy to see that $\mathrm{E}(X_1) = \mathrm{E}(X_2)$. But to work out the value you need to make some clever use of geometric series or some calculus. In the end $\mathrm{E}(X_i) = \frac{1}{.25} = 4$, so $\mathrm{E}(X) = 8$.

9. A bowl of M&M's is almost empty. There are 3 reds, 5 blues, and 8 browns. You randomly select three.

a. What is the probability that you get one of each color?
b. What is the probability that they are all brown?
c. What is the probability that they are all red?
d. What is the probability that they are all the same color?
e. If they are all the same color, what is the probability that they are all brown?
f. What is the probability that you get 2 browns and one blue?
g. What is the probability that you get two of one color and one of a different color?
h. What is the expected number of brown candies you select?

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***Solution.***

| part | answer |
|---|---|
| a | 0.214285714 |
| b | 0.100000000 |
| c | 0.001785714 |
| d | 0.119642857 |
| e | 0.098073555 |
| f | 0.250000000 |
| g | 0.666071429 |
| h | 1.500000000 |

For part a) $\frac{\binom{8}{1}\binom{5}{1}\binom{3}{1}}{\binom{16}{3}}$.
In h) split the variable into three parts $X = X_1 + X_2 + X_3$ where $X_i$ is 1 if candy $i$ is brown, else it is 0. Then $E(X_i) = 0.5$ and the sum is 1.5.

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